Havalyn Hale

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EDUCATION

B.S. in Electrical and Computer Engineering *Olin College of Engineering* Expected May 2029

Needham, MA

4-Year 50% Merit Tuition Scholarship. GPA: 4.0. Relevant Coursework: Software Design, Discrete Mathematics (in progress), Modeling and Simulation, Quantitative Engineering Analysis, Instrumentation Sensors and Measurement

EXPERIENCE

CNH May 2027 - August 2027

Electrical Design Engineering Intern (Incoming) *Sioux Falls, SD*

- Joining the electrical hardware team responsible for PCB-based electronic hardware design, taking circuits from schematic to board in Altium Designer
- Team builds systems around Linux-based processors and microcontrollers, including HMI devices, communication gateways, and GNSS receivers
- Role covers circuit validation through simulation and proof-of-concept prototypes, hardware validation, and schematic and layout reviews

Olin Electric Motorsports September 2026 - Present

Low Voltage Engineer *Needham, MA*

- Finishing the Formula SAE car's telemetry system: Python on a Raspberry Pi (Linux) relaying CAN bus data to a ground station over 915 MHz SiK radios
- Reconfigured the radios from 9600 to 57600 baud and redesigned the message format, cutting each packet from ~1,870 bytes to ~40 and raising throughput from ~3 to 100+ messages/sec; added per-ID rate limiting
- Tested the original 9600 baud link at 151 ft with no line of sight: 4.8% message loss over 500 messages (target less than 10%)

Olin Vine Robotics August 2026 - Present

Research Assistant *Needham, MA*

- Designed a steering control board with 6 independent pressure channels (regulator, gauge, pressure sensor, and vent valve each), giving the robot 6 DOF
- Wired a Raspberry Pi Pico sensing board that reads a Honeywell ABP sensor's raw 14-bit output over I²C; wrote embedded C firmware that rejects bad readings and zeroes the sensor at startup, bringing noise down to a few thousandths of a psi
- Building a second sensing board that reads 6 Honeywell ABP 0-5 psi sensors through a multiplexer across three robot sections

Olin Shop August 2026 - Present

Electronics Assistant *Needham, MA*

- Redesigned soldering training to add de-soldering; 3D printed cases for 2 Saleae logic analyzers (SOLIDWORKS); re-drew the training schematic in KiCAD

Olin Plasma Engineering Lab (OPEL) January 2026 - Present

Researcher *Needham, MA*

- Co-developed MATLAB models for the MK5 Hall Effect Thruster on an 8 kg CubeSat: sized propellant (0.11-0.22 kg for a 280 m/s apogee raise) and predicted approximately 15 mN thrust, 1,630 s Isp, and 305 W; compiled results as an Olin ISR
- Co-developed design requirements for a 3U CubeSat-scale thruster; performed thin-wall stress analysis of an Al 6061-T6 propellant tank (5 cm radius, 3 mm wall, 14.3 MPa), yielding a 1.08 safety factor

PROJECTS

Ground Station RF Link Model (Olin Electric Motorsports)

Worked the link budget by hand, then implemented it in Python with NumPy; modeled a 1 km, 915 MHz link (27 dBm transmit, -91.7 dB free-space loss, -12.7 dB ground reflection) predicting -75.4 dBm received power and a 29.6 dB margin against a 20 dB target

Python, NumPy

Introduction to Radio Frequency Parameters (QEA, self-selected topic)

Formalized Z, Y, ABCD, and S-parameter representations and derived relationships between them, including cascaded ABCD matrices; applied Smith Chart analysis to impedance matching with complex port impedances; presented to 50+ students

Rocky Animatronic (PIE, team project)

4-legged walking robot with 12 servos; wired and labeled the servo driver and ran a test crawl gait to characterize power draw and servo heating (2.05 A at 6 V, 12.3 W)

C++, Arduino Nano

SKILLS

Programming: Python (NumPy), C, C++, MATLAB, Java

Embedded & Systems: Linux/Ubuntu, Embedded Systems, Raspberry Pi, CAN Bus, Serial Communication (pyserial), I²C, PWM, Arduino, RF Link Budgets

Tools & Lab: Git/GitHub, Altium Designer, KiCAD, SOLIDWORKS, COMSOL, Oscilloscope, Logic Analyzer, Soldering/De-soldering, 3D Printing